

VIT CHENNAI · BMAT101L

Calculus and Its Applications

Complete Study Guide · 7 Modules · Formula Sheets · Solved Problems

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Module 2: Multivariable Calculus

Module 3: Multiple Integrals

Module 4: Beta & Gamma

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Mean Value Theorems & Taylor Series

The Mean Value Theorems connect the derivative of a function to its average rate of change. Taylor and Maclaurin series allow any smooth function to be represented as an infinite polynomial — fundamental for approximations and error analysis.

1.1 Rolle's Theorem & Lagrange's MVT

ROLLE'S THEOREM

If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then $\exists c \in (a, b)$ such that $f'(c) = 0$.

LAGRANGE'S MVT (LMVT)

If f is continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

CAUCHY'S MVT (CMVT)

If f, g continuous on $[a, b]$, differentiable on (a, b) , $g'(x) \neq 0$, then $\exists c \in (a, b)$ such that:

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

1 Verify Rolle's Theorem

Verify Rolle's theorem for $f(x) = x^2 - 5x + 6$ on $[2, 3]$.

SOLUTION

- 1 f is polynomial $\rightarrow$ continuous and differentiable everywhere $\checkmark$
- 2 $f(2) = 4 - 10 + 6 = 0$ and $f(3) = 9 - 15 + 6 = 0$. So $f(2) = f(3)$ $\checkmark$

3 $f'(x) = 2x - 5 = 0 \Rightarrow x = 2.5$. Since $2.5 \in (2, 3)$, Rolle's theorem is verified ✓

$$c = 2.5 \in (2, 3)$$

2 Apply LMVT

Find c for $f(x) = x^3 - x^2 - x + 1$ on $[0, 2]$.

SOLUTION

1 $f(0) = 1, f(2) = 8 - 4 - 2 + 1 = 3$. Slope $= \frac{3-1}{2-0} = 1$

2 $f'(x) = 3x^2 - 2x - 1 = 1 \Rightarrow 3x^2 - 2x - 2 = 0 \Rightarrow x = \frac{2 \pm \sqrt{4+24}}{6} = \frac{2 \pm \sqrt{28}}{6} \approx \frac{2 \pm 5.29}{6}$

3 $x \approx 1.215$ (taking +) or $x \approx -0.548$ (rejected — not in $(0,2)$). So $c \approx 1.215$ ✓

$$c \approx 1.215$$

1.2 Taylor's & Maclaurin's Series

TAYLOR / MACLAURIN FORMULAS

TAYLOR'S SERIES ABOUT $x=a$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

MACLAURIN SERIES ($A = 0$)

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \dots$$

E^x

$$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

SIN x

$$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

COS x

$$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

LN(1+x)

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \dots, |x| \leq 1$$

1 Maclaurin expansion

Expand $f(x) = e^x \sin x$ up to x^4 using Maclaurin series.

STEP-BY-STEP SOLUTION

$$1 \quad e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots$$

$$2 \quad \sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$$

$$3 \quad \text{Multiply term by term up to } x^4: \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}\right)\left(x - \frac{x^3}{6}\right)$$

$$4 \quad \text{Collect: } x + x^2 + \frac{x^3}{2} + \frac{x^4}{6} - \frac{x^3}{6} - \frac{x^4}{6} \dots = x + x^2 + \frac{x^3}{3} - \frac{x^4}{30} + \dots$$

$$e^x \sin x = x + x^2 + x^3/3 - x^4/30 + \dots$$

2 Approximate using Taylor

Approximate $\cos(0.2)$ using first three non-zero terms of Maclaurin series.

SOLUTION

$$1 \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$2 \quad \cos(0.2) \approx 1 - \frac{(0.2)^2}{2} + \frac{(0.2)^4}{24} = 1 - \frac{0.04}{2} + \frac{0.0016}{24} = 1 - 0.02 + 0.0000667$$

$$\cos(0.2) \approx 0.9800667$$

Module 1 — Key Formulas

LMVT

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Taylor about a

$$\sum \frac{f^{(n)}(a)}{n!} (x - a)^n$$

e^x Maclaurin

$$1 + x + x^2/2! + x^3/3! + \dots$$

Functions of several variables require partial differentiation. Euler's theorem, Jacobians, and Lagrange multipliers extend single-variable techniques to handle physical problems involving multiple interacting quantities.

2.1 Partial Derivatives

PARTIAL DERIVATIVE RULES

DEFINITION

$$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$$

CLAIRAUT'S THEOREM

$$f_{xy} = f_{yx} \text{ (if continuous)}$$

TOTAL DIFFERENTIAL

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

CHAIN RULE (COMPOSITE)

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

IMPLICIT (F(X,Y,Z)=0)

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

NOTATION

$$f_{xx} = \partial^2 f / \partial x^2, \quad f_{xy} = \partial^2 f / \partial x \partial y$$

1 Second-order partial derivatives

If $u = x^3y^2 + 3xy^3$, find u_{xx}, u_{yy}, u_{xy} .

SOLUTION

$$1 \quad u_x = 3x^2y^2 + 3y^3 \Rightarrow u_{xx} = 6xy^2$$

$$2 \quad u_y = 2x^3y + 9xy^2 \Rightarrow u_{yy} = 2x^3 + 18xy$$

$$3 \quad u_{xy} = \frac{\partial}{\partial y}(3x^2y^2 + 3y^3) = 6x^2y + 9y^2$$

$$u_{xx}=6xy^2, u_{yy}=2x^3+18xy, u_{xy}=6x^2y+9y^2$$

2.2 Euler's Theorem for Homogeneous Functions

EULER'S THEOREM

If $f(x, y)$ is homogeneous of degree n , then:

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f$$

Corollary: $x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy} = n(n-1)f$

1 Apply Euler's Theorem

If $u = \sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, show that $xu_x + yu_y = \frac{1}{2}\tan u$.

SOLUTION

- 1 Let $f = \sin u = \frac{x+y}{\sqrt{x}+\sqrt{y}}$. Check degree: $f(tx, ty) = \frac{t(x+y)}{\sqrt{tx}+\sqrt{ty}} = \frac{t^{1/2}(x+y)}{t^{1/2}(\sqrt{x}+\sqrt{y})} = t^{1/2}f$. So f is homogeneous of degree $n=1/2$.
- 2 By Euler's: $x \frac{\partial(\sin u)}{\partial x} + y \frac{\partial(\sin u)}{\partial y} = \frac{1}{2} \sin u$
- 3 $\cos u (xu_x + yu_y) = \frac{1}{2} \sin u \Rightarrow xu_x + yu_y = \frac{1}{2} \tan u \checkmark$

$xu_x + yu_y = \frac{1}{2} \tan u$ (proved)

2.3 Jacobians

JACOBIAN DETERMINANT

JACOBIAN $J = \partial(U, V) / \partial(X, Y)$

$$J = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

INVERSE RULE

$J \cdot J' = 1$ where $J' = \partial(x, y) / \partial(u, v)$

1 Jacobian of polar coordinates

Find $\partial(x, y) / \partial(r, \theta)$ where $x = r \cos \theta$, $y = r \sin \theta$.

SOLUTION

$$1 \quad J = \begin{vmatrix} x_r & x_\theta \\ y_r & y_\theta \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$2 \quad = r \cos^2 \theta + r \sin^2 \theta = r$$

J = r (hence dA = r dr dθ in polar)

2.4 Maxima, Minima & Saddle Points

SECOND DERIVATIVE TEST

CRITICAL POINTS

Solve $f_x = 0$ and $f_y = 0$ simultaneously

DISCRIMINANT D

$$D = f_{xx}f_{yy} - (f_{xy})^2$$

CLASSIFICATION

$D > 0, f_{xx} > 0 \rightarrow \text{Min}; D > 0, f_{xx} < 0 \rightarrow \text{Max};$
 $D < 0 \rightarrow \text{Saddle}; D = 0 \rightarrow \text{test fails}$

1 Find extrema

Find extrema of $f(x, y) = x^3 + 3xy^2 - 15x - 12y$.

SOLUTION

$$1 \quad f_x = 3x^2 + 3y^2 - 15 = 0 \Rightarrow x^2 + y^2 = 5$$

$$2 \quad f_y = 6xy - 12 = 0 \Rightarrow xy = 2 \Rightarrow y = 2/x$$

$$3 \quad \text{Substitute: } x^2 + 4/x^2 = 5 \Rightarrow x^4 - 5x^2 + 4 = 0 \Rightarrow (x^2 - 1)(x^2 - 4) = 0. \text{ So } x = \pm 1, \pm 2.$$

$$4 \quad \text{Points: } (1, 2), (-1, -2), (2, 1), (-2, -1). \text{ Compute } D \text{ at each: } f_{xx} = 6x, f_{yy} = 6x, f_{xy} = 6y. D = (6x)(6x) - (6y)^2 = 36(x^2 - y^2).$$

$$5 \quad \text{At } (2, 1): D = 36(4 - 1) > 0, f_{xx} = 12 > 0 \rightarrow \text{local min. At } (-2, -1): D > 0, f_{xx} < 0 \rightarrow \text{local max.}$$

Local min at (2,1); Local max at (-2,-1)

2.5 Lagrange Multipliers

LAGRANGE'S METHOD

To extremise $f(x, y, z)$ subject to $g(x, y, z) = 0$: solve $\nabla f = \lambda \nabla g$ simultaneously with the constraint.

1 Lagrange multipliers

Find the maximum of $f = xyz$ subject to $x + y + z = a$ ($x, y, z > 0$).

SOLUTION

- 1 $\nabla f = \lambda \nabla g \Rightarrow yz = \lambda, xz = \lambda, xy = \lambda$
- 2 From equations: $yz = xz \Rightarrow x = y$ and $xz = xy \Rightarrow y = z$. So $x = y = z$.
- 3 Constraint: $3x = a \Rightarrow x = y = z = a/3$. Max value: $f = (a/3)^3 = a^3/27$

Maximum of $xyz = a^3/27$ at $x=y=z=a/3$

Module 2 — Key Formulas

Euler's Theorem

$$xf_x + yf_y = nf$$

Discriminant D

$$f_{xx}f_{yy} - (f_{xy})^2$$

Polar Jacobian

$$\partial(x, y)/\partial(r, \theta) = r$$

Multiple Integrals

Double and triple integrals extend single integration to higher dimensions, computing areas, volumes, masses, and moments of inertia. Changing variables (especially to polar/cylindrical/spherical coordinates) simplifies many integrals.

3.1 Double Integrals — Cartesian

DOUBLE INTEGRAL RULES

ITERATED INTEGRAL

$$\iint_R f \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx$$

CHANGE OF ORDER

Draw the region, re-express limits with y as outer variable

AREA

$$A = \iint_R dA$$

1 Evaluate a double integral

Evaluate $\int_0^1 \int_0^x (x^2 + y^2) \, dy \, dx$.

SOLUTION

1 Inner integral: $\int_0^x (x^2 + y^2) \, dy = \left[x^2 y + \frac{y^3}{3} \right]_0^x = x^3 + \frac{x^3}{3} = \frac{4x^3}{3}$

2 Outer: $\int_0^1 \frac{4x^3}{3} \, dx = \frac{4}{3} \cdot \frac{x^4}{4} \Big|_0^1 = \frac{1}{3}$

1/3

2 Change of order of integration

Change the order and evaluate $\int_0^1 \int_x^1 e^{y^2} \, dy \, dx$.

SOLUTION

1 Region: $0 \leq x \leq 1$, $x \leq y \leq 1$. In reversed order: $0 \leq y \leq 1$, $0 \leq x \leq y$

2
$$\int_0^1 \int_0^y e^{y^2} dx dy = \int_0^1 e^{y^2} \cdot y dy = \frac{1}{2} \left[e^{y^2} \right]_0^1 = \frac{e-1}{2}$$

$(e-1)/2$

3.2 Double Integrals in Polar Coordinates

POLAR COORDINATE INTEGRAL

SUBSTITUTION

$x = r \cos \theta$, $y = r \sin \theta$, $dA = r dr d\theta$

CIRCLE $X^2 + Y^2 = A^2$

r ranges 0 to a ; θ ranges 0 to 2π

1 Polar integral for area

Find the area enclosed by $r = a(1 + \cos \theta)$ (cardioid).

SOLUTION

1
$$A = \frac{1}{2} \int_0^{2\pi} r^2 d\theta = \frac{1}{2} \int_0^{2\pi} a^2 (1 + \cos \theta)^2 d\theta$$

2
$$= \frac{a^2}{2} \int_0^{2\pi} (1 + 2 \cos \theta + \cos^2 \theta) d\theta = \frac{a^2}{2} [2\pi + 0 + \pi] = \frac{3\pi a^2}{2}$$

$A = 3\pi a^2/2$

3.3 Triple Integrals & Applications

TRIPLE INTEGRAL APPLICATIONS

VOLUME

CYLINDRICAL COORDINATES

SPHERICAL COORDINATES

$$V = \iiint_E dV$$

$$dV = r \, dr \, d\theta \, dz$$

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

1 Volume by triple integral

Find the volume bounded by $z = x^2 + y^2$ and $z = 4$.

SOLUTION

1 Intersection: $x^2 + y^2 = 4$, i.e., $r=2$ in polar. Switch to cylindrical: $V = \int_0^{2\pi} \int_0^2 \int_{r^2}^4 r \, dz \, dr \, d\theta$

2 Inner z-integral: $r(4 - r^2)$. Radial: $\int_0^2 r(4 - r^2) \, dr = [2r^2 - r^4/4]_0^2 = 8 - 4 = 4$

3 $V = \int_0^{2\pi} 4 \, d\theta = 8\pi$

$$V = 8\pi$$

Module 3 — Key Formulas

Polar Area element

$$dA = r \, dr \, d\theta$$

Cylindrical dV

$$r \, dr \, d\theta \, dz$$

Change order tip

Sketch region, swap dy dx

Beta and Gamma functions evaluate integrals involving exponentials and powers that arise throughout physics and engineering. Gamma generalises the factorial; Beta links to probability distributions and appears in evaluating definite integrals.

4.1 Gamma Function

GAMMA FUNCTION

DEFINITION

$$\Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx, \quad n > 0$$

RECURRENCE

$$\Gamma(n+1) = n\Gamma(n); \quad \Gamma(n+1) = n! \text{ for integer } n$$

SPECIAL VALUES

$$\Gamma(1) = 1; \quad \Gamma(1/2) = \sqrt{\pi}; \quad \Gamma(3/2) = \sqrt{\pi}/2$$

1 Evaluate using Gamma

Evaluate $\int_0^{\infty} x^4 e^{-3x} dx$.

SOLUTION

1 Let $t = 3x \Rightarrow x = t/3, dx = dt/3$. Integral becomes: $\int_0^{\infty} (t/3)^4 e^{-t} \frac{dt}{3} = \frac{1}{3^5} \int_0^{\infty} t^4 e^{-t} dt$

2 $= \frac{\Gamma(5)}{3^5} = \frac{4!}{243} = \frac{24}{243} = \frac{8}{81}$

8/81

2 Compute $\Gamma(7/2)$

Find $\Gamma(7/2)$.

SOLUTION

1 $\Gamma(7/2) = \frac{5}{2}\Gamma(5/2) = \frac{5}{2} \cdot \frac{3}{2}\Gamma(3/2) = \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2}\Gamma(1/2) = \frac{15}{8}\sqrt{\pi}$

$$\Gamma(7/2) = 15\sqrt{\pi}/8$$

4.2 Beta Function

BETA FUNCTION

DEFINITION

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

RELATION TO GAMMA

$$B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

TRIG FORM

$$B(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$$

1 Evaluate with Beta function

Evaluate $\int_0^{\pi/2} \sin^5 \theta \cos^3 \theta d\theta$.

SOLUTION

1 Use trig form: $= \frac{1}{2} B(3, 2) = \frac{1}{2} \cdot \frac{\Gamma(3)\Gamma(2)}{\Gamma(5)} = \frac{1}{2} \cdot \frac{2! \cdot 1!}{4!} = \frac{1}{2} \cdot \frac{2}{24} = \frac{1}{24}$

$$1/24$$

Module 4 — Key Formulas

$$\Gamma(n+1)$$

$$n\Gamma(n)$$

$$B(m, n)$$

$$\Gamma(m)\Gamma(n)/\Gamma(m+n)$$

$$\Gamma(1/2)$$

$$\sqrt{\pi}$$

Vector calculus unifies differentiation and integration in three dimensions. The gradient points in the direction of steepest ascent; divergence and curl characterise flux and rotation; the integral theorems of Green, Stokes, and Gauss link these operations beautifully.

5.1 Gradient, Divergence & Curl

DIFFERENTIAL OPERATORS

DEL OPERATOR ∇

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

GRADIENT (SCALAR $\rightarrow$ VECTOR)

$$\text{grad } f = \nabla f = f_x \mathbf{i} + f_y \mathbf{j} + f_z \mathbf{k}$$

DIVERGENCE (VECTOR $\rightarrow$ SCALAR)

$$\text{div } \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

CURL (VECTOR $\rightarrow$ VECTOR)

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ P_x & P_y & P_z \\ Q_x & Q_y & Q_z \end{vmatrix}$$

SOLENOIDAL FIELD

$$\nabla \cdot \mathbf{F} = 0$$

IRROTATIONAL FIELD

$$\nabla \times \mathbf{F} = \mathbf{0}$$

1 Directional derivative

Find the directional derivative of $\phi = xy^2 + yz^3$ at $(1, -1, 1)$ in direction of $\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.

SOLUTION

1 $\nabla \phi = (y^2)\mathbf{i} + (2xy + z^3)\mathbf{j} + (3yz^2)\mathbf{k}$. At $(1, -1, 1)$: $\nabla \phi = \mathbf{i} + (-2 + 1)\mathbf{j} + (-3)\mathbf{k} = \mathbf{i} - \mathbf{j} - 3\mathbf{k}$

2 Unit vector: $\hat{n} = \frac{\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}}{3}$

3 $D_{\hat{n}}\phi = \nabla \phi \cdot \hat{n} = \frac{1 - 2 - 6}{3} = -\frac{7}{3}$

Directional derivative = $-7/3$

5.2 Green's, Stokes' & Gauss Theorems

INTEGRAL THEOREMS

GREEN'S THEOREM (2D)

$$\oint_C (P dx + Q dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

STOKES' THEOREM (3D)

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$$

GAUSS DIVERGENCE THEOREM

$$\oiint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{F}) dV$$

1 Apply Green's Theorem

Evaluate $\oint_C (y dx - x dy)$ where C is the circle $x^2 + y^2 = 4$ (counterclockwise).

SOLUTION

1 $P = y, Q = -x. \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -1 - 1 = -2$

2 By Green's: $\iint_R (-2) dA = -2 \cdot \pi(2)^2 = -8\pi$

-8π

2 Apply Gauss Divergence Theorem

Evaluate $\oiint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ over the sphere $x^2 + y^2 + z^2 = a^2$.

SOLUTION

1 $\nabla \cdot \mathbf{F} = 1 + 1 + 1 = 3$

2 By Gauss: $\iiint_V 3 dV = 3 \cdot \frac{4}{3}\pi a^3 = 4\pi a^3$

$4\pi a^3$

Gradient

$$\nabla f = f_x \mathbf{i} + f_y \mathbf{j} + f_z \mathbf{k}$$

Green's Theorem

$$\oint P dx + Q dy = \iint (Q_x - P_y) dA$$

Gauss Theorem

$$\text{Vol} \nabla \cdot \mathbf{F} = \iint \mathbf{F} \cdot d\mathbf{S} = \iiint \nabla \cdot \mathbf{F} dV$$

Sequences are ordered lists of numbers converging to a limit. Series are sums of sequences. Convergence tests determine whether infinite sums converge to finite values — critical for validating Taylor series, Fourier series, and numerical approximations.

6.1 Sequences — Convergence

SEQUENCE TESTS

CONVERGENCE

$\lim_{n \rightarrow \infty} a_n = L$ (finite) $\rightarrow$ converges to L

MONOTONE CONVERGENCE THM

Monotone + bounded $\rightarrow$ converges

SANDWICH THEOREM

If $b_n \leq a_n \leq c_n$ and $b_n, c_n \rightarrow L$, then $a_n \rightarrow L$

1 Convergence of a sequence

Show that $a_n = \frac{3n^2 + 5}{2n^2 - 1}$ converges and find its limit.

SOLUTION

- 1 Divide numerator and denominator by n^2 : $a_n = \frac{3 + 5/n^2}{2 - 1/n^2}$
- 2 As $n \rightarrow \infty$: numerator $\rightarrow 3$, denominator $\rightarrow 2$. So $a_n \rightarrow 3/2$

Sequence converges to $3/2$

6.2 Tests for Series Convergence

CONVERGENCE TESTS

COMPARISON TEST

RATIO TEST (D'ALEMBERT)

ROOT TEST (CAUCHY)

If $0 \leq a_n \leq b_n$: $\sum b_n$ converges $\Rightarrow \sum a_n$ converges

$L = \lim \left| \frac{a_{n+1}}{a_n} \right|$: $L < 1$ conv; $L > 1$ div; $L = 1$ inconclusive

$L = \lim |a_n|^{1/n}$: $L < 1$ conv; $L > 1$ div

INTEGRAL TEST

$\sum f(n)$ and $\int_1^\infty f(x)dx$ both converge or both diverge

ALTERNATING SERIES TEST

$\sum (-1)^n a_n$ converges if a_n decreasing and $a_n \rightarrow 0$

P-SERIES

$\sum 1/n^p$ converges iff $p > 1$

1 Ratio test

Test convergence of $\sum_{n=1}^{\infty} \frac{n!}{2^n}$.

SOLUTION

1 $\frac{a_{n+1}}{a_n} = \frac{(n+1)!/2^{n+1}}{n!/2^n} = \frac{n+1}{2}$

2 $L = \lim_{n \rightarrow \infty} \frac{n+1}{2} = \infty > 1 \Rightarrow$ Series **diverges**

Diverges (ratio $\rightarrow \infty$)

2 Root test

Test $\sum_{n=1}^{\infty} \left(\frac{n}{n+1} \right)^{n^2}$.

SOLUTION

1 $L = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n = \lim \left(1 - \frac{1}{n+1} \right)^n = e^{-1} = \frac{1}{e} < 1$

2 Since $L = 1/e < 1$, series **converges**

Converges ($L = 1/e < 1$)

6.3 Power Series & Radius of Convergence

POWER SERIES

POWER SERIES FORM

$\sum_{n=0}^{\infty} c_n (x - a)^n$ converges in $\{ |x - a|$

RADIUS OF CONVERGENCE

$$R = \lim \left| \frac{c_n}{c_{n+1}} \right| \text{ or } R = 1 / \limsup |c_n|^{1/n}$$

INTERVAL OF CONVERGENCE

Check endpoints separately for absolute convergence

1 Find radius of convergence

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n \cdot 3^n}$.

SOLUTION

1 Apply ratio test: $\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{x^{n+1}}{(n+1)3^{n+1}} \cdot \frac{n \cdot 3^n}{x^n} \right| = \frac{n}{n+1} \cdot \frac{|x|}{3} \rightarrow \frac{|x|}{3}$

2 Converges when $|x|/3 < 1 \Rightarrow |x| < 3$. So $R = 3$

$$R = 3$$

Module 6 — Key Tests

Ratio Test L

$$\lim |a_{n+1}/a_n|: <1 \text{ conv, } >1 \text{ div}$$

Root Test L

$$\lim |a_n|^{1/n}: <1 \text{ conv, } >1 \text{ div}$$

p-series

$$\sum 1/n^p \text{ conv iff } p > 1$$

When exact solutions are impossible, numerical methods give highly accurate approximations. Root-finding locates zeros of functions; numerical integration computes definite integrals with controllable error; finite differences interpolate and differentiate tabulated data.

7.1 Root Finding Methods

ROOT FINDING

BISECTION METHOD

If $f(a)f(b) < 0$, root in $[a, b]$. New midpoint $c = (a+b)/2$; repeat on half containing sign change

NEWTON-RAPHSON

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

CONVERGENCE

Bisection: linear; Newton-Raphson: quadratic (very fast near root)

1 Newton-Raphson method

Find a root of $f(x) = x^3 - x - 1 = 0$ starting from $x_0 = 1.5$ (2 iterations).

STEP-BY-STEP SOLUTION

1 $f'(x) = 3x^2 - 1$

2 Iter 1: $f(1.5) = 3.375 - 1.5 - 1 = 0.875$; $f'(1.5) = 5.75$. $x_1 = 1.5 - \frac{0.875}{5.75} = 1.5 - 0.1522 = 1.3478$

3 Iter 2: $f(1.3478) = 2.4448 - 1.3478 - 1 = 0.0970$; $f'(1.3478) = 4.4463$. $x_2 = 1.3478 - \frac{0.0970}{4.4463} = 1.3260$

Root ≈ 1.3260 (true root ≈ 1.3247)

7.2 Numerical Integration

NUMERICAL INTEGRATION FORMULAS

TRAPEZOIDAL RULE

$$\int_a^b f dx \approx \frac{h}{2} [y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n], \quad h =$$

SIMPSON'S 1/3 RULE (N EVEN)

$$\approx \frac{h}{3} [y_0 + 4y_1 + 2y_2 + 4y_3 + \cdots + 4y_{n-1} + y_n]$$

ERROR — TRAPEZOIDAL

$$E \approx -\frac{(b-a)h^2}{12} f''(\xi)$$

1 Simpson's 1/3 rule

Evaluate $\int_0^1 \frac{1}{1+x} dx$ using Simpson's rule with $n=4$.

STEP-BY-STEP SOLUTION

1 $h=0.25$. Values: $y_0 = f(0) = 1$, $y_1 = f(0.25) = 0.8$, $y_2 = f(0.5) = 0.667$, $y_3 = f(0.75) = 0.571$, $y_4 = f(1) = 0.5$

2 Simpson's: $\frac{0.25}{3} [1 + 4(0.8) + 2(0.667) + 4(0.571) + 0.5] = \frac{0.25}{3} [1 + 3.2 + 1.333 + 2.286 + 0.5]$

3 $= \frac{0.25}{3} [8.319] = \frac{2.0798}{3} = 0.6933$

4 Exact: $\ln 2 = 0.6931$. Error = 0.0002 (excellent!)

≈ 0.6933 (exact: $\ln 2 = 0.6931$)

7.3 Finite Differences & Interpolation

FINITE DIFFERENCE OPERATORS

FORWARD DIFFERENCE Δ

$$\Delta y_n = y_{n+1} - y_n; \quad \Delta^2 y_n = \Delta y_{n+1} - \Delta y_n$$

BACKWARD DIFFERENCE ∇

$$\nabla y_n = y_n - y_{n-1}$$

NEWTON'S FORWARD INTERPOLATION

$$y = y_0 + s\Delta y_0 + \frac{s(s-1)}{2!}\Delta^2 y_0 + \cdots \text{ where}$$

$$s = \frac{x-x_0}{h}$$

1 Forward difference table

Construct the forward difference table for f : $x=0 \rightarrow 1, f=1$; $x=1 \rightarrow 2, f=8$; $x=2 \rightarrow 3, f=27$; $x=3 \rightarrow 4, f=64$.

SOLUTION

- 1 Δy : $8-1=7$, $27-8=19$, $64-27=37$
- 2 $\Delta^2 y$: $19-7=12$, $37-19=18$
- 3 $\Delta^3 y$: $18-12=6$. $\Delta^4 y=0$. (Note: for $f(x)=x^3$, $\Delta^3 f=3!=6$ ✓)

$\Delta^3 y = 6$ (constant for cubic polynomial)

Module 7 — Key Formulas

Newton-Raphson

$$x_{n+1} = x_n - f(x_n)/f'(x_n)$$

Trapezoidal

$$\frac{h}{2} [y_0 + 2 \sum y_i + y_n]$$

Simpson's 1/3

$$\frac{h}{3} [y_0 + 4y_1 + 2y_2 + \cdots + y_n]$$

BMAT101L — Master Formula Sheet

Modules 1–2: Calculus & Multivariable

LMVT $f'(c) = (f(b) - f(a))/(b - a)$	TAYLOR $\sum f^{(n)}(a)(x - a)^n/n!$	EULER THEOREM $xf_x + yf_y = nf$
DISCRIMINANT D $f_{xx}f_{yy} - (f_{xy})^2$	JACOBIAN (POLAR) r	LAGRANGE $\nabla F = \lambda \nabla G$ solve with constraint

Modules 3–4: Integrals & Special Functions

POLAR DA $r \, dr \, d\theta$	$\Gamma(N+1)$ $n\Gamma(n) = n!$	B(M,N) $\Gamma(m)\Gamma(n)/\Gamma(m + n)$
$\Gamma(1/2)$ $\sqrt{\pi}$	CARDIOID AREA $3\pi a^2/2$	CYLINDRICAL DV $r \, dr \, d\theta \, dz$

Modules 5–7: Vector Calc, Series & Numerics

GREEN'S THEOREM $\oint Pdx + Qdy = \iint (Q_x - P_y)dA$	GAUSS THEOREM $\oiint \mathbf{F} \cdot d\mathbf{S} = \iiint \nabla \cdot \mathbf{F} \, dV$	RATIO TEST $L < 1$ conv; $L > 1$ div
NEWTON-RAPHSON $x_{n+1} = x_n - f/f'$	SIMPSON'S 1/3 $h/3[y_0 + 4y_1 + 2y_2 + \dots]$	FORWARD DIFF Δ $\Delta y_n = y_{n+1} - y_n$

BMAT101L — Complete

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